

# GlobalFlow Under Pressure: Network Design, Disruption, and Resilience in a Multi-Commodity Logistics Network

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**Abstract**—This paper applies a Multi-Commodity Capacitated Fixed-Charge Network Design (MC-FCND) framework to GlobalFlow, a third-party logistics operator routing three product families — fertilizers, semiconductors, and battery components — across a four-tier hierarchical network of 59 nodes and 317 arcs. Phase 1 formulates and solves the baseline MIP exactly using Python + Xpress, compares four formulations (F0–F3) on LP relaxation quality, and identifies the optimal configuration ( $Z^* = 4,454,801\text{€}$ , 14 of 15 warehouses open, 11 of 47 fixed arcs activated). Phase 2 subjects this configuration to six adversarial scenarios and one free scenario, evaluating three response strategies (rerouting, adaptation, full redesign). Results show that F0, combining disaggregated warehouse linking with aggregated arc capacity, produces the tightest LP relaxation (gap 0.5343%). The network exhibits a critical structural vulnerability: Singapore hub concentration drives 82.09% of the cumulative disruption cost ( $\Sigma\Delta Z = 9,657,154\text{€}$ ), with the combined shock S3 alone reaching  $\Delta Z = +118.78\%$ . Pre-contracting four backup arcs to Dubai at 2,500€ each yields a robustified network at a resilience premium of +0.224%, reducing the cumulative disruption burden by −72.9% and delivering an aggregate ROI of 704.3%.

**Index Terms**—multi-commodity flow, fixed-charge network design, facility location, MIP formulation, LP relaxation, supply chain resilience, disruption scenarios, hub concentration, GlobalFlow, FICO Xpress

## I INTRODUCTION

The optimal design of a logistics network under normal conditions does not guarantee its robustness against the geopolitical and economic disruptions that characterize today’s global trade environment. This tension between optimality and resilience constitutes the central question of this work, applied to the case of GlobalFlow, a third-party logistics operator managing the worldwide distribution of three product families with distinct risk profiles.

**Context.** GlobalFlow operates on a four-tier hierarchical network: nine product-specialized suppliers (S), five intercontinental hubs (H), fifteen regional warehouses whose opening is decided strategically (W), and thirty final customers whose demand must be compulsorily satisfied (C). The three products  $P = \{A, B, C\}$  — fertilizers, semiconductors, and battery components — share the network infrastructure but exhibit radically different cost structures, preferred transport modes,

and exposures to exogenous shocks. The geopolitical environment of recent years makes this challenge particularly acute: a surge in energy costs linked to the Russo-Ukrainian and Iranian conflicts, disruption of semiconductor supply chains by the Taiwan Strait crisis, South Korean export restrictions, and the reimposition of American tariffs on transatlantic and transpacific corridors.

**Objectives.** This report pursues two complementary objectives. In Phase 1, we formulate GlobalFlow’s problem as a MIP of the MC-FCND class, solve the baseline instance exactly using Python+Xpress, and analyze the quality of four alternative formulations according to their LP relaxation bounds. In Phase 2, we subject the optimal configuration obtained to six adversarial scenarios and one free scenario, and determine for each the optimal response strategy among rerouting, adaptation, and full redesign.

**Research questions.** This work answers five precise questions:

- **RQ1:** Which MIP formulation produces the tightest LP relaxation, and why?
- **RQ2:** What is the optimal baseline network configuration  $W^*$ ,  $A^*$ , and what is the total cost  $Z^*$  decomposed into its fixed and variable components?
- **RQ3:** Among the six adversarial scenarios, which is the most devastating for GlobalFlow, and what are its structural causes?
- **RQ4:** Does the baseline network exhibit recurrent structural vulnerabilities — hub concentration, critical arc dependency, or warehouse over-opening — that compound disruption costs across scenarios?
- **RQ5:** What is the cost of resilience — how much more would a robustified network cost at baseline to significantly reduce disruption costs  $\Delta Z$ ?

**Report structure.** Section 2 presents the complete MIP formulation and the analysis of variants F0–F3. Section 3 reports the optimal results of Phase 1. Section 4 describes the methodological framework for stress-testing. Section 5

analyzes the Phase 2 results for each scenario. Section 6 synthesizes the structural vulnerabilities, evaluates the resilience premium, and answers the five research questions. Section 7 concludes with limitations and perspectives.

## II MIP FORMULATION – MODEL & VARIANTS

### II-A Problem Identification

The problem analyzed belongs to the class of Multi-Commodity Capacitated Fixed-Charge Network Design (MC-FCND) with facility location. Formally, it consists in simultaneously routing  $|P| = 3$  commodities over a directed graph  $G = (N, A)$ , deciding which facilities (warehouses) to open and which fixed-cost arcs to activate, so as to minimize the total cost while exactly satisfying each customer's demand for each product.

The network comprises  $|N| = 59$  nodes distributed across four disjoint sets: suppliers  $S$  ( $|S| = 9$ ), hubs  $H$  ( $|H| = 5$ ), warehouses  $W$  ( $|W| = 15$ ), and customers  $C$  ( $|C| = 30$ ), with  $N = S \cup H \cup W \cup C$ . The arc set  $A$  ( $|A| = 317$ ) partitions into  $A_F$  ( $|A_F| = 47$  arcs with fixed activation cost) and  $A_0 = A \setminus A_F$  (always available arcs, primarily supplier-to-hub links).

### II-B Notation and Sets

The sets, parameters, and variables used across all formulations are defined below in a unified manner.

#### Indices and sets.

|                              |                                                                                                                          |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| $p \in P = \{A, B, C\}$      | the products ( $A$ = Fertilizers, $B$ = Semiconductors, $C$ = Battery Components); $ P  = 3$ .                           |
| $i, j \in N$                 | the network nodes; $ N  = 59$ .                                                                                          |
| $i \in S \subset N$          | the suppliers; $ S  = 9$ .                                                                                               |
| $i \in H \subset N$          | the international hubs, always active; $ H  = 5$ . $H = \{\text{Frankfurt, Dubai, Singapore, Chicago, São Paulo}\}$ .    |
| $i \in W \subset N$          | the regional warehouses, whose opening is to be decided; $ W  = 15$ .                                                    |
| $i \in C \subset N$          | the customers (sinks); $ C  = 30$ .                                                                                      |
| $S, H, W, C$                 | pairwise disjoint sets; $N = S \cup H \cup W \cup C$ .                                                                   |
| $(i, j) \in A$               | the directed arcs of the network; $ A  = 317$ .                                                                          |
| $(i, j) \in A_F \subseteq A$ | arcs with fixed activation cost (fixed_activation_cost > 0); $ A_F  = 47$ .                                              |
| $A_0 = A \setminus A_F$      | always available arcs (supplier → hub, zero fixed cost).                                                                 |
| $i \in S^p \subseteq S$      | suppliers dedicated to product $p$ . $S^A = \{S_1, S_2, S_3\}$ , $S^B = \{S_4, S_5, S_6\}$ , $S^C = \{S_7, S_8, S_9\}$ . |
| $\delta^+(i), \delta^-(i)$   | for every node $i$ : $\delta^+(i) = \{j : (i, j) \in A\}$ the set of successors                                          |

and  $\delta^-(i) = \{j : (j, i) \in A\}$  the set of predecessors.

#### Parameters.

|                       |                                                                                                                                                                                                           |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Supply $_i^p$         | supply capacity of product $p$ at supplier $i \in S^p$ (units per period).                                                                                                                                |
| Demand $_i^p$         | demand for product $p$ at customer $i \in C$ (units per period).                                                                                                                                          |
| $c_{ij}^p$            | variable transportation cost per unit of product $p$ on arc $(i, j) \in A$ , including interzonal tariff surcharges $\tau_{z_f, z_t} \cdot c_{ij, \text{eff}}^p = c_{ij}^p \cdot (1 + \tau_{z_f, z_t})$ . |
| $f_{ij}^{\text{arc}}$ | fixed activation cost of arc $(i, j) \in A_F$ .                                                                                                                                                           |
| $f_i^{\text{wh}}$     | fixed opening cost of warehouse $i \in W$ .                                                                                                                                                               |
| $u_{ij}$              | shared capacity of arc $(i, j)$ across all products.                                                                                                                                                      |
| CapW $_i$             | physical capacity of warehouse $i \in W$ (total inbound flow, all products combined).                                                                                                                     |

#### Decision variables.

|                       |                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------|
| $x_{ij}^p \geq 0$     | flow of product $p$ on arc $(i, j) \in A$ — continuous operational routing variable. |
| $y_i \in \{0, 1\}$    | equals 1 if warehouse $i \in W$ is open — strategic binary location variable.        |
| $z_{ij} \in \{0, 1\}$ | equals 1 if arc $(i, j) \in A_F$ is activated — network design binary variable.      |

### II-C Reference Formulation F0

Formulation F0 constitutes the reference model submitted to the solver. It rests on two modelling choices: a disaggregated per-arc warehouse linking constraint (strong Axis W) and an aggregated arc capacity constraint over products (standard Axis A).

#### II-C1 Objective function (F0-0)

The objective is to minimize the total network cost, composed of the fixed costs of warehouse opening, the fixed costs of arc activation, and the variable transportation costs:

$$\min Z = \underbrace{\sum_{i \in W} f_i^{\text{wh}} \cdot y_i}_{\text{(I) warehouse opening}} + \underbrace{\sum_{(i, j) \in A_F} f_{ij}^{\text{arc}} \cdot z_{ij}}_{\text{(II) arc activation}} + \underbrace{\sum_{p \in P} \sum_{(i, j) \in A} c_{ij}^p \cdot x_{ij}^p}_{\text{(III) variable transport}} \quad (\text{F0-0})$$

#### II-C2 Supplier supply (F0-1)

Each supplier  $i \in S^p$  may ship in total at most up to its own production capacity:

$$\sum_{j: (i, j) \in A} x_{ij}^p \leq \text{Supply}_i^p \quad \forall p \in P, \forall i \in S^p \quad (\text{F0-1})$$

This constraint is formulated individually per supplier (and not aggregated over  $S^p$ ) to respect industrial realism: Taipei ( $S_4$ ) cannot ship more than 741 units independently of Seoul's or Yokohama's production.

### II-C3 Product specialization of suppliers (F0-1b)

A supplier dedicated to product  $p$  injects no other product into the network:

$$x_{ij}^{p'} = 0 \quad \forall p \in P, \forall i \in S^p, \forall j \in \delta^+(i), \forall p' \neq p \quad (\text{F0-1b})$$

On the GlobalFlow instance, this constraint is satisfied implicitly: each supplier appears for only one product in the Supply table, so that  $\text{Supply}_i^{p'} = 0$  for all  $p' \neq p$ , and (F0-1) already bounds these flows to zero. It is nonetheless included to guarantee the formal consistency of the model.

### II-C4 Flow conservation — hubs and warehouses (F0-2)

Each hub and each warehouse is a pure transit node: for each product, the inbound flow equals the outbound flow, with no creation or destruction:

$$\sum_{j \in \delta^-(i)} x_{ji}^p = \sum_{j \in \delta^+(i)} x_{ij}^p \quad \forall p \in P, \forall i \in H \cup W \quad (\text{F0-2})$$

### II-C5 Customer demand satisfaction (F0-3)

Each customer's demand is exactly satisfied for each product:

$$\sum_{j \in \delta^-(i)} x_{ji}^p = \text{Demand}_i^p \quad \forall p \in P, \forall i \in C \quad (\text{F0-3})$$

### II-C6 Warehouse linking constraint — disaggregated per arc (F0-4a/4b)

This is the structural constraint of formulation F0. It imposes that if warehouse  $i$  is closed ( $y_i = 0$ ), no flow may transit through any of its incident arcs — neither inbound nor outbound:

$$\sum_{p \in P} x_{ji}^p \leq u_{ji} \cdot y_i \quad \forall i \in W, \forall j \in \delta^-(i) \quad (\text{F0-4a})$$

$$\sum_{p \in P} x_{ij}^p \leq u_{ij} \cdot y_i \quad \forall i \in W, \forall j \in \delta^+(i) \quad (\text{F0-4b})$$

The disaggregated form (one constraint per incident arc) is strictly stronger than the aggregated form of F1 in the LP relaxation. When  $y_i = \alpha \in (0, 1)$ , each arc is individually bounded to  $u \cdot \alpha$ , which prevents the LP solver from using an arc at full capacity while keeping the warehouse only partially open. For example, for  $y_w = 0.5$  and an inbound arc of capacity  $u = 300$ : F0 bounds this flow to 150 (50%), while F1 allows it to reach 300 (100%). This property is the analogue of the strong constraint  $x_{ij} \leq y_i$  in the classical Facility Location problem.

### II-C7 Warehouse capacity (F0-5)

The total inbound flow, across all products, is bounded by the physical capacity of the warehouse:

$$\sum_{p \in P} \sum_{j \in \delta^-(i)} x_{ji}^p \leq \text{CapW}_i \cdot y_i \quad \forall i \in W \quad (\text{F0-5})$$

This constraint is active only when the sum of the capacities of the inbound arcs exceeds  $\text{CapW}_i$ ; otherwise, it is dominated by constraints (F0-4a).

### II-C8 Shared arc capacity (F0-6a/6b)

For fixed-cost arcs, activation  $z_{ij} = 0$  prevents any flow;  $z_{ij} = 1$  allows a total flow (all products) at most equal to the shared capacity  $u_{ij}$ . The natural big-M  $u_{ij}$  is the tightest possible bound in aggregated form:

$$\sum_{p \in P} x_{ij}^p \leq u_{ij} \cdot z_{ij} \quad \forall (i, j) \in A_F \quad (\text{F0-6a})$$

For free arcs (supplier  $\rightarrow$  hub, zero fixed cost), the capacity applies directly without a binary variable:

$$\sum_{p \in P} x_{ij}^p \leq u_{ij} \quad \forall (i, j) \in A_0 \quad (\text{F0-6b})$$

### II-C9 Variable domains (F0-7/8/9)

$$x_{ij}^p \geq 0 \quad \forall p \in P, (i, j) \in A \quad (\text{F0-7})$$

$$y_i \in \{0, 1\} \quad \forall i \in W \quad (\text{F0-8})$$

$$z_{ij} \in \{0, 1\} \quad \forall (i, j) \in A_F \quad (\text{F0-9})$$

## II-D Variants F1, F2, and F3

The four formulations result from the combination of two independent modelling axes. The integer optimal solution  $Z^*$  is identical for all four models; only the quality of the LP relaxation differs.

**Formulation F1 — aggregated linking.** F1 replaces constraints (F0-4a) and (F0-4b) with a single global constraint per warehouse, which simultaneously plays the role of linking constraint and capacity bound:

$$\sum_{p \in P} \sum_{j \in \delta^-(i)} x_{ji}^p \leq \text{CapW}_i \cdot y_i \quad \forall i \in W \quad (\text{F1-4})$$

This formulation is more compact, but produces a weaker LP relaxation: with  $y_i = \alpha \in (0, 1)$ , the solver may saturate any individual inbound arc at its full capacity  $u_{ji}$ , provided the global sum remains below  $\text{CapW}_i \cdot \alpha$ . Constraint (F0-5) becomes redundant in F1 and is removed.

**Formulation F2 — arc capacity disaggregated per product.** F2 replaces the aggregated constraint (F0-6a) with an individual per-product constraint on each fixed-cost arc:

$$x_{ij}^p \leq u_{ij} \cdot z_{ij} \quad \forall p \in P, \forall (i, j) \in A_F \quad (\text{F2-6a})$$

This constraint is weaker than (F0-6a) in the LP relaxation: summing (F2-6a) over  $p \in P$  gives  $\sum_p x_{ij}^p \leq |P| \cdot u_{ij} \cdot z_{ij}$ , which is strictly more permissive than (F0-6a). With  $z_{ij} = 0.5$  for example, the aggregated form caps the *total* flow across all products at  $u_{ij}/2$ , while the disaggregated form allows each product individually up to  $u_{ij}/2$ , for a potential combined

flow of  $|P| \cdot u_{ij}/2 = 3u_{ij}/2$ . Axis A disaggregation therefore relaxes the arc capacity constraint in the LP relaxation. F2 nonetheless outperforms F1 because its strong Axis W disaggregated warehouse linking more than compensates for this relaxation on this instance.

**Formulation F3 — mixed.** F3 combines the aggregated linking of F1 (weak Axis W) with the disaggregated capacity of F2 (weak Axis A). It illustrates the interaction of the two axes: Axis W disaggregation tightens the LP bound, while Axis A disaggregation loosens it. F3 is weaker than F1 on Axis A — the per-product disaggregation allows a combined arc flow of  $|P| \cdot u_{ij} \cdot z_{ij}$  instead of  $u_{ij} \cdot z_{ij}$  — and equally weak on Axis W. It thus cumulates both relaxations, producing the lowest LP bound of the four formulations.

### II-E LP Relaxation Analysis and Choice of F0

The four formulations produce the same integer optimal solution  $Z^* = 4,454,800.57$  €, but differ in the quality of their linear relaxation. Two independent axes distinguish them: Axis W (warehouse linking disaggregated per arc in F0/F2, aggregated in F1/F3) and Axis A (fixed arc capacity aggregated over products in F0/F1, disaggregated per product in F2/F3).

**Theoretical ordering.** The two modelling axes have opposite effects on LP relaxation quality: Axis W disaggregation tightens the bound, while Axis A disaggregation loosens it. This establishes the theoretical ordering:

$$Z_{LP}^*(F3) \leq Z_{LP}^*(F1) \leq Z_{LP}^*(F2) \leq Z_{LP}^*(F0)$$

F3 is weakest (both axes loose) and F0 is tightest (strong Axis W combined with tight Axis A).

**Numerical results.** The LP–IP gap is defined by  $\text{gap}(F) = (Z^* - Z_{LP}^*(F))/Z^* \times 100\%$ .

**Table 1:** LP relaxation bounds and LP–IP gaps (baseline instance).

| F                        | Axis W  | Axis A  | LP Bound (€) | Gap (%) |
|--------------------------|---------|---------|--------------|---------|
| F0                       | Disagg. | Agg.    | 4,430,996.94 | 0.5343  |
| F2                       | Disagg. | Disagg. | 4,427,994.32 | 0.6017  |
| F1                       | Agg.    | Agg.    | 4,424,642.73 | 0.6770  |
| F3                       | Agg.    | Disagg. | 4,420,984.56 | 0.7591  |
| $Z^*$ (all formulations) |         |         | 4,454,800.57 | —       |

The numerical results (Table 1) confirm the theoretical ordering  $Z_{LP}^*(F3) \leq Z_{LP}^*(F1) \leq Z_{LP}^*(F2) \leq Z_{LP}^*(F0)$ . Axis W disaggregation raises the LP bound ( $F0 > F1$ ,  $F2 > F3$ ), while Axis A disaggregation lowers it ( $F1 > F3$ ,  $F0 > F2$ ). The result  $F3 < F1$  is therefore not counter-intuitive but expected: F3 cumulates both relaxations — weak Axis W and loose Axis A — and produces the lowest bound of the four formulations.

**Choice of F0.** F0 produces the tightest LP bound (gap 0.5343%) because it combines the strongest constraint on each axis: disaggregated warehouse linking (tight Axis W) and aggregated arc capacity (tight Axis A). F2, despite sharing the strong Axis W of F0, is weakened by its looser Axis A and produces a larger gap (0.6017%). F0 therefore constitutes the

best formulation both theoretically and numerically on this instance.

## III PHASE 1 – BASELINE OPTIMAL SOLUTION

### III-A Prior Corrections to the Instance

Two inconsistencies in the database used (`globalflow_instance.xlsx`) required corrections before solving, applied directly in the data loading code.

**Under-dimensioned arc capacities.** Two free arcs had a declared capacity lower than the optimal flow they must carry, making them infeasible without correction:

- A008 (Yokohama  $S_6 \rightarrow$  Singapore  $H_3$ , product B): capacity corrected to 741 units, corresponding exactly to Yokohama’s supply. The arc being the only outgoing path from this supplier, its capacity must at minimum equal the available supply.
- A011 (Chengdu  $S_9 \rightarrow$  Singapore  $H_3$ , product C): capacity corrected to 631 units, corresponding to Chengdu’s supply for Battery Components.

**Supply/demand imbalance.** The supply of S6 (Yokohama, product: Semiconductors) is reduced from 741 to 740 units. Without this correction, the total supply of semiconductors exceeds the aggregate demand by 1 unit, which, combined with the exact flow conservation constraints (F0-2) and demand satisfaction constraints (F0-3), renders the model infeasible: the surplus unit can neither be absorbed by a customer (exact demand) nor stored (warehouses are pure transit nodes).

### III-B Optimal Configuration $W^*$ and $A^*$

**Open warehouses.** Out of 15 candidate warehouses, 14 are open in the optimal solution. Only W9 (Sydney) remains closed, explained by the absence of Australian demand in the instance and by an opening cost not justified by the flows that would pass through it.

Utilization rates are highly heterogeneous: Rotterdam (W1) is saturated at 91.3%, Bangalore (W15) at 76.5%, and Manaus (W14) at 62.5%, while Chennai (W5) is used at only 4.1% and Monterrey (W13) at 8.7%. Although lightly loaded, these warehouses remain open because they are the only available service points for certain customers with no routing alternative (customers C30 Kolkata via W5, and C8 Calgary via W13).

**Activated fixed arcs.** Only 11 of the 47 fixed arcs are activated in  $A^*$ , a 23.4% activation rate. These 11 arcs all correspond to last-mile warehouse–customer links, concentrated in North America (4 arcs), South America/Brazil (2 arcs), and Asia-Pacific (5 arcs). No fixed hub–warehouse arc is activated, meaning the hub-to-warehouse network relies exclusively on variable-cost arcs.

### III-C Decomposition of the Optimal Cost $Z^*$

The cost structure (Table 2) is dominated at 99.1% by variable transport, with fixed costs representing less than 1% of the total. Interzonal tariff surcharges account for 3.3% of total cost, i.e. 149 136 €, across 525 of the 951 arc-product pairs (55% carry at least one interzonal surcharge).



**Table 2:** Cost decomposition of the optimal solution  $Z^*$ .

| Component                          | Cost (€)          | Share (%)    |
|------------------------------------|-------------------|--------------|
| Warehouse opening                  | 22 046.49         | 0.5          |
| Fixed arc activation               | 19 638.46         | 0.4          |
| <i>Subtotal fixed costs</i>        | <i>41684.95</i>   | <i>0.9</i>   |
| Variable transport (base)          | 4 263 979.55      | 95.7         |
| Interzonal tariff surcharges       | 149 136.07        | 3.3          |
| <i>Subtotal variable transport</i> | <i>4413115.62</i> | <i>99.1</i>  |
| <b>Total <math>Z^*</math></b>      | <b>4454800.57</b> | <b>100.0</b> |

The per-product decomposition (Table 3) reveals a striking heterogeneity:

**Table 3:** Variable cost decomposition by product.

| Product                | Flow (units) | Flow %       | Var. cost (€)     | Cost %       |
|------------------------|--------------|--------------|-------------------|--------------|
| A — Fertilizers        | 6272         | 28.8         | 638 198.25        | 14.5         |
| B — Semiconductors     | 9585         | 44.0         | 2 814 741.04      | 63.8         |
| C — Battery Components | 5765         | 26.5         | 960 176.34        | 21.8         |
| <b>Total</b>           | <b>21622</b> | <b>100.0</b> | <b>4413115.62</b> | <b>100.0</b> |

The Semiconductors accounts for 44% of total flow but concentrates 63.8% of variable cost. This over-representation is explained by structurally higher unit variable costs on the corridors serving this product — notably the long routes from Singapore to Asian and North-American warehouses, which combine large distances and Asia↔Americas interzonal tariff surcharges among the highest in the instance. Product A presents conversely a cost share (14.5%) well below its flow share (28.8%), consistent with predominantly European corridors with low tariff surcharges. Battery Components presents a cost share (21.8%) slightly above its flow share (26.5%), indicating a moderate overrepresentation consistent with unit costs higher than Product A but substantially lower than Product B.

### III-D Routing Plan – Flow Structure

The optimal routing is organized into seven hierarchical link types. Table 4 summarizes the variable cost subtotals.

**Table 4:** Variable cost by link type.

| Link type                      | Var. cost (€)  | Share (%)    |
|--------------------------------|----------------|--------------|
| Suppliers → Hubs               | 1 256 824      | 28.5         |
| Hubs → Hubs                    | 179 192        | 4.1          |
| Hubs → Warehouses              | 1 586 254      | 35.9         |
| Warehouses → Hubs              | 656 082        | 14.9         |
| Warehouses → Warehouses        | 6569           | 0.1          |
| Warehouses → Customers (free)  | 634 377        | 14.4         |
| Warehouses → Customers (fixed) | 93 818         | 2.1          |
| <b>Total</b>                   | <b>4413115</b> | <b>100.0</b> |

**Suppliers → Hubs.** All 9 suppliers route their entire production to their entry hubs. Frankfurt (H1) absorbs fertilizer production from Casablanca (S2, 694 units) and a partial flow from Minsk (S1, 356 units); the remaining Minsk production (338 units) is routed to Dubai (H2). Singapore (H3) concentrates virtually all semiconductor flow — Taipei (S4, 741), Seoul (S5, 741), Yokohama (S6, 740) — as well as Battery

Components from Chengdu (S9, 631). São Paulo (H5) receives Port of Spain (S3, 537 units of fertilizers) and Antofagasta (S7, 631 units of Battery Components). Dubai (H2) receives Lubumbashi (S8, 631 units of Battery Components) and a partial flow from Minsk (S1, 338 units of fertilizers). Port of Spain also routes 157 units of fertilizers to Chicago (H4).

**Hubs → Hubs.** A single inter-hub flow is observed in the optimal solution: Dubai (H2) transfers to Frankfurt (H1) 403 units of semiconductors and 80 units of Battery Components, for an effective cost of 179 192 € (Table 4). This transit results from the combined pressure of MiddleEast↔Europe tariff surcharges and the capacity structure of Dubai–warehouse arcs, which makes it more economical to consolidate these flows via Frankfurt before redistribution to European warehouses.

**Hubs → Warehouses.** This is the most expensive level of the network (35.9% of total variable cost). Frankfurt feeds 7 warehouses (Rotterdam, Madrid, Warsaw, Osaka, Toronto, Montreal, Manaus). Singapore redistributes to 6 warehouses (Mumbai, Chennai, Osaka, Tokyo, Shenzhen, Bangalore). Dubai and São Paulo each serve 5 warehouses: respectively (Rotterdam, Madrid, Warsaw, Mumbai, Bangalore) and (Houston, Toronto, Montreal, Monterrey, Manaus). Chicago is limited to 3 warehouses (Houston, Toronto, Monterrey).

**Warehouses → Hubs.** Three return warehouse–hub flows are activated, all carrying exclusively semiconductors, for a total cost of 656 082 € (Table 4, 14.9% of variable cost). These are: Mumbai (W4) → Dubai (H2, 349 units), Bangalore (W15) → Dubai (H2, 373 units), and Osaka (W6) → Chicago (H4, 536 units). The latter arc (A133) is by far the most expensive in the network at 513 824 € alone, reflecting both the Asia↔Americas distance and the associated tariff surcharge. These return flows reflect a double-loop organization: Singapore distributes semiconductors to Asian warehouses, which then redistribute to American hubs for final delivery.

**Warehouses → Warehouses.** Only three inter-warehouse flows are activated, of marginal volume (6 569 € in total, 0.1% of variable cost): Madrid (W2) → Montreal (W12) for 6 units of Battery Components, and Mumbai (W4) → Tokyo (W7) and Shenzhen (W8) for 13 units each of fertilizers. These flows correspond to local capacity adjustments to balance Japanese fertilizer demand that hubs cannot serve directly in an economical way.

**Warehouses → Customers.** Final delivery is carried out via 92 distinct flows to the 30 customers, split between free arcs (634 377 €) and activated fixed arcs (93 818 €). Notably, three fixed arcs — A288 (Tokyo → TokyoCity), A291 (Osaka → OsakaCity), and A309 (Mumbai → MumbaiCity) — display a near-zero effective variable cost (25 €, 0 €, and 0 € respectively for all three products combined), meaning their activation is justified solely by the contractual fixed cost, the warehouses being geographically adjacent to the customers they serve. Similarly, arc A312 (Bangalore → BangaloreCity) has zero variable cost for all three products, Bangalore being both the warehouse and the customer city.

### III-E Commentary on the Structure of the Solution

The optimal solution reveals four key structural characteristics.

**Near-total warehouse opening.** The opening of 14 out of 15 warehouses is explained by the low weight of fixed costs (0.9% of  $Z^*$ ): as soon as a warehouse is the only local service point, opening it is systematically preferable to the rerouting surcharge. Chennai (4.1%) and Monterrey (8.7%) are a direct illustration of this.

**Critical concentration on Singapore.** Singapore is the sole entry hub for all semiconductors and a significant share of Battery Components. This structural dependency is the network's primary vulnerability — its closure instantly deprives all Asian warehouses of their supply.

**A costly Asian double loop.** Arc A133 (Osaka → Chicago) alone represents 11.6% of total variable cost (513 824 €). This double-loop organization — Singapore feeds Asian warehouses, which redistribute to Chicago — results from the absence of a sufficiently capacitated direct arc between Singapore and North America.

**Negligible fixed costs as a signal for Phase 2.** With only 41 685 € in fixed costs, response strategies to adverse scenarios will not be able to rely on significant fixed-cost savings — the challenge will almost exclusively be to contain the increase in variable costs.

## IV STRESS-TESTING FRAMEWORK

### IV-A Objective

The optimal baseline configuration  $G^* = (N, A^*)$  obtained in Phase 1 is subjected to a structured battery of six adversarial scenarios and one free scenario. For each scenario, the objective is to determine the response strategy that minimizes total cost under the new conditions.

### IV-B Response Strategies

For each scenario, three strategies are evaluated and compared.

**Strategy R — Rerouting only.** The Phase 1 configuration ( $W^*, A^*$ ) is fixed on its still-operational assets in the scenario. Only flows are re-optimized. Fixed costs paid correspond solely to assets that remain operational — a warehouse or arc from  $W^*/A^*$  removed by the scenario is not counted. The total comparable cost is:

$$\text{Total}_R = \text{sunk}_{W^* \cap \text{op}} + \text{sunk}_{A^* \cap \text{op}} + Z_{\text{var}}^{\text{re-opt}}$$

**Strategy A — Adaptation.** A partial reconfiguration from the Phase 1 solution is allowed: opening or closing warehouses, activating or deactivating arcs. This strategy does not start from scratch — it starts from the current configuration ( $W^*, A^*$ ) and allows adjustments by paying the associated costs. A closing penalty equal to the opening cost is applied for each warehouse in  $W^*$  that is still operational but closed in the new configuration, and a deactivation penalty equal to the

fixed cost is applied for each arc in  $A^*$  that is still operational but deactivated. The total comparable cost is:

$$\text{Total}_A = \text{fix}_W^{\text{new}} + \text{fix}_A^{\text{new}} + Z_{\text{var}} + \sum_{\substack{w \in W^* \cap \text{op} \\ w \text{ closed}}} f_w^{\text{wh}} + \sum_{\substack{a \in A^* \cap \text{op} \\ a \text{ deactivated}}} f_a^{\text{arc}}$$

**Strategy F — Full redesign.** The Phase 1 model is solved from scratch with the scenario's parameters. No constraints are inherited from Phase 1, no transition penalties apply. This strategy provides the theoretical lower bound on achievable cost under the scenario, but ignores sunk and transition costs. In general,  $\text{Total}_F \leq \text{Total}_A$ .

### IV-C Adversarial Scenarios

The seven scenarios are summarized in Table 5.

**Table 5: Summary of adversarial scenarios.**

| ID | Type               | Description                                                                                                                                                  | Products |
|----|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| T1 | Tariff shock       | Trump-era tariffs +40% on transatlantic and transpacific corridors                                                                                           | B, C     |
| T2 | Energy crisis      | Asymmetric cost increase driven by Ukraine/Iran conflict: air freight +60%, sea freight +10%, road +20%                                                      | A, B, C  |
| T3 | Seasonal surcharge | Hub handling fees +30% on all hub nodes during Q4                                                                                                            | A, B     |
| S1 | Node disruption    | Closure of Singapore hub (H3), simulating Taiwan Strait crisis                                                                                               | B, C     |
| S2 | Arc disruption     | Blockade of the Korea→Europe export corridor (South Korean export restrictions)                                                                              | B        |
| S3 | Combined shock     | Simultaneous application of T2 and S1                                                                                                                        | A, B, C  |
| S4 | Free scenario      | Persian Gulf / Strait of Hormuz crisis: removal of 14 maritime arcs H2↔Americas, Suez surcharge +25%, war-risk premium +40% on remaining active H2 corridors | A, B, C  |

For scenarios S1 and S3, the closure of Singapore removes 42 arcs from the network and deprives Asian suppliers of their sole entry hub. To guarantee model feasibility, four emergency arcs are created towards Dubai (H2): Taipei → H2, Seoul → H2, Yokohama → H2, and Chengdu → H2, with a cost multiplier of  $4.5\times$  relative to the corresponding baseline arcs, reflecting the recourse to emergency air freight towards a distant alternative hub (the only transport mode fast enough to maintain flow continuity for semiconductors and Battery Components).

Scenario S4 is the group's free scenario. It models a military escalation in the Persian Gulf in 2025: the Iranian navy and Houthis extend their operations into the Gulf of Oman, blocking Dubai's outgoing maritime routes towards the Americas. Dubai remains operational for Europe and Asia, but with a war-risk premium of +40% on these active corridors. A persistent Suez surcharge of +25% is applied to all canal-dependent maritime arcs. All customers remain reachable via alternative hubs H1, H3, H4, and H5.

### IV-D Implementation

All scenarios are solved to near-exact optimality (MIP gap  $\leq 0.0013\%$  for Phase 1,  $\leq 0.0001\%$  for Phase 2), in full consistency with the Phase 1 methodology. The model used is formulation F0 for all three strategies. Phase 1 results ( $W^*, A^*, Z^*$ ) are loaded from the JSON file exported in Phase 1, guaranteeing total consistency between the two phases.

## V PHASE 2 – SCENARIO ANALYSIS

The optimal baseline configuration  $G^* = (N, A^*)$  is subjected here to seven adversarial scenarios, each evaluated under three response strategies (F, A, R) as defined in Section IV. All 21 sub-problems are solved to exact optimality, in full methodological consistency with Phase 1.

### V-A Summary of Results

**Table 6: Phase 2 results summary**

| ID                                                                | Best | $\Delta Z$ (€)    | $\Delta Z$ (%) | WH open (F) |
|-------------------------------------------------------------------|------|-------------------|----------------|-------------|
| S3                                                                | F    | +5,291,249        | +118.78%       | 11          |
| S1                                                                | F    | +2,636,979        | +59.19%        | 11          |
| T3                                                                | F    | +852,284          | +19.13%        | 14          |
| T2                                                                | F    | +474,045          | +10.64%        | 14          |
| S4                                                                | F    | +247,141          | +5.55%         | 14          |
| T1                                                                | F    | +155,456          | +3.49%         | 14          |
| S2                                                                | F    | $\approx 0$       | 0.00%          | 14          |
| <b>Cumulative <math>\Sigma \Delta Z</math> (optimal strategy)</b> |      | <b>+9,657,154</b> | —              |             |

Table 6 reports, for each scenario, the optimal strategy, the disruption cost  $\Delta Z$ , and the number of open warehouses under the Full Redesign solution. Full Redesign (F) is optimal or co-optimal in every case without exception. This result is mathematically expected, since F constitutes the theoretical lower bound on achievable cost by ignoring all sunk and transition expenditures. The practical value of the comparison therefore lies in quantifying the penalty of operational inertia: how much more does GlobalFlow pay if it cannot reconfigure the network (R), or can only partially adapt it (A)?

The seven scenarios produce a cumulative disruption cost of  $\Sigma \Delta Z = 9,657,154$  € under the optimal strategy. The distribution is highly asymmetric: scenarios S3 and S1 — both involving the closure of Singapore hub (H3) — jointly account for 82.0% of the total, while S2 registers a disruption cost of precisely zero. Scenarios S1 and S3 both require the activation of four emergency airlift arcs towards Dubai (departing from Taipei, Seoul, Yokohama and Chengdu) created to restore feasibility after the removal of Singapore, at a cost multiplier of  $4.5\times$  the corresponding baseline corridor cost.

**Table 7: Total comparable cost by strategy and scenario**

| Scenario | F — Total (€)    | A — Total (€) | R — Total (€) | Best           |
|----------|------------------|---------------|---------------|----------------|
| T1       | <b>4,610,256</b> | 4,610,341     | 4,610,797     | F              |
| T2       | <b>4,928,846</b> | 4,928,846     | 4,928,846     | F = A = R      |
| T3       | <b>5,307,084</b> | 5,307,901     | 5,406,145     | F              |
| S1       | <b>7,091,780</b> | 7,101,753     | 7,111,826     | F              |
| S2       | <b>4,454,801</b> | 4,454,801     | 4,454,801     | F = A = R = Z* |
| S3       | <b>9,746,049</b> | 9,755,073     | 9,769,939     | F              |
| S4       | <b>4,701,942</b> | 4,701,942     | 4,723,597     | F = A          |

Table 7 presents the full F/A/R comparison by scenario. The theoretical hierarchy  $\text{Total}_F \leq \text{Total}_A \leq \text{Total}_R$  is verified in every instance without exception, which validates the internal consistency of the three strategy formulations. Two structural observations stand out. First, in T2 and S2, all three strategies yield identical totals, a degenerate case explained in

Section V-B. Second, the marginal advantage of Adaptation over Rerouting is significant only in T3 ( $\Delta = 98,243$ €,  $-1.82\%$ ) and essentially negligible in T1, T2, S1, and S4. This reflects the near-total opening of warehouses in the baseline ( $|W^*| = 14/15$ ), which leaves little structural slack for reconfiguration.

### V-B Per-Scenario Analysis

Table 8 provides the cost decomposition of the optimal (Full Redesign) solution for each scenario, broken into fixed warehouse costs, fixed arc activation costs, and variable transportation costs as well as the variable cost share by product.

**Table 8: Cost decomposition of the Full Redesign solution by scenario. All monetary values in €**

| Sc.      | Fix. WH | Fix. arc | Variable  | Variable cost share |       |       | Total     | $\Delta Z\%$  |
|----------|---------|----------|-----------|---------------------|-------|-------|-----------|---------------|
|          |         |          |           | A (%)               | B (%) | C (%) |           |               |
| Baseline | 22,046  | 19,638   | 4,413,116 | 14.5                | 63.8  | 21.8  | 4,454,801 | —             |
| T1       | 22,046  | 21,222   | 4,566,988 | 14.9                | 63.8  | 21.3  | 4,610,256 | +3.49%        |
| T2       | 22,046  | 19,638   | 4,887,161 | 14.6                | 63.6  | 21.8  | 4,928,846 | +10.64%       |
| T3       | 22,046  | 31,119   | 5,253,919 | 14.1                | 63.1  | 22.7  | 5,307,084 | +19.13%       |
| S1       | 15,915  | 16,546   | 7,059,318 | 9.0                 | 71.7  | 19.3  | 7,091,780 | +59.19%       |
| S2       | 22,046  | 19,638   | 4,413,116 | 14.5                | 63.8  | 21.8  | 4,454,801 | $\approx 0\%$ |
| S3       | 15,915  | 18,098   | 9,712,036 | 7.4                 | 75.0  | 17.7  | 9,746,049 | +118.78%      |
| S4       | 22,046  | 22,309   | 4,657,586 | 14.5                | 62.8  | 22.7  | 4,701,942 | +5.55%        |

Tables 9–15 detail strategy results across scenarios and baseline comparison. Scenario-specific analyses are provided below each table.

**Table 9: Strategy comparison — T1 (Tariff shock, +40% trans-corridors).**

| Strategy | Total (€)        | $\Delta Z$ (€)  | $\Delta Z$ (%) | Prem. vs F (€) |
|----------|------------------|-----------------|----------------|----------------|
| <b>F</b> | <b>4,610,256</b> | <b>+155,456</b> | <b>+3.49%</b>  | —              |
| A        | 4,610,341        | +155,541        | +3.49%         | +85            |
| R        | 4,610,797        | +155,997        | +3.50%         | +541           |

**T1 — Tariff shock ( $\Delta Z = +3.49\%$ ).** The reimposition of transatlantic and transpacific tariffs at +40% raises total cost by 155,456€, the smallest impact among cost-shock scenarios. The response is essentially a flow redistribution within an unchanged network topology: the Full Redesign activates two additional fixed arcs (A188, Rotterdam-Montreal, and A224, Montreal-New York, compared to  $A^*$ ), and remove one arc (A225, Toronto-New York) to exploit an alternative. But it opens the same 14 warehouses as the baseline. Semiconductors bear the largest absolute increase in variable cost (+96,575€, +3.4%), driven by transpacific semiconductor flows, battery components follow (+13,896€). The Adaptation and Rerouting strategies differ from F by only 85€ and 541€ respectively, confirming that the network topology is already near-optimal for this cost structure and that no structural reconfiguration yields meaningful savings.

**Table 10: Strategy comparison — T2 (Energy crisis, air +60%, road +20%, sea +10%).**

| Strategy | Total (€) | $\Delta Z$ (€) | $\Delta Z$ (%) | Prem. vs F (€) |
|----------|-----------|----------------|----------------|----------------|
| F        | 4,928,846 | +474,045       | +10.64%        | —              |
| A        | 4,928,846 | +474,045       | +10.64%        | 0              |
| R        | 4,928,846 | +474,045       | +10.64%        | 0              |

F=A=R: reconfiguration yields no benefit.

**T2 — Energy crisis** ( $\Delta Z = +10.64\%$ ). The asymmetric energy shock (air freight +60%, road +20%, sea +10%) raises total cost by 474,045€, borne almost entirely by variable transport. Since air travel is excluded from our data, rising airfare does not affect our model. The three strategies produce identical totals (4,928,846€), a degenerate outcome arising from the fact that the optimal network topology under T2 is exactly the baseline ( $W^*, A^*$ ). Full Redesign rediscovers the same 14 warehouses and 11 fixed arcs, so its fixed cost equals the sunk cost embedded in R and the penalty-free reopening cost of A. The cost increase is therefore structural, no reconfiguration can absorb it.

**Table 11: Strategy comparison — T3 (Seasonal hub surcharge, +30%).**

| Strategy | Total (€) | $\Delta Z$ (€) | $\Delta Z$ (%) | Prem. vs F (€) |
|----------|-----------|----------------|----------------|----------------|
| F        | 5,307,084 | +852,284       | +19.13%        | —              |
| A        | 5,307,901 | +853,101       | +19.15%        | +817           |
| R        | 5,406,145 | +951,344       | +21.35%        | +99,060        |

**T3 — Seasonal hub surcharge** ( $\Delta Z = +19.13\%$ ). The uniform +30% handling fee on all five hubs during Q4 delivers the largest disruption cost among the purely cost-based scenarios, surpassing the energy crisis T2 by 378,238€. The explanation is structural: every unit of flow in the network, regardless of product or corridor, must transit through at least one hub. The surcharge is therefore multiplicative across the entire volume. The Asian double-loop further amplifies the damage with Osaka-Chicago arc (A133) carrying alone  $\approx 536$  units of semiconductors. The Fixed arc activation cost rises from the baseline (19,638€ to 31,119€) because Full Redesign activates 10 additional arcs while removing the Toronto-New York arc. Here, F outperforms R by 99,060€ (−1.87%), the largest operational flexibility premium of any single-node cost scenario.

**Table 12: Strategy comparison — S1 (Singapore hub closure).**

| Strategy | Total (€) | $\Delta Z$ (€) | $\Delta Z$ (%) | Prem. vs F (€) |
|----------|-----------|----------------|----------------|----------------|
| F        | 7,091,780 | +2,636,979     | +59.19%        | —              |
| A        | 7,101,753 | +2,646,952     | +59.42%        | +9,973         |
| R        | 7,111,826 | +2,657,025     | +59.64%        | +20,046        |

**S1 — Singapore hub closure** ( $\Delta Z = +59.19\%$ ). The simulated Taiwan Strait crisis is by far the most severe single disruption: removing Singapore hub eliminates several arcs from the network, severs all three semiconductor suppliers

(Taipei, Seoul, Yokohama) and Chengdu's battery component supply from their sole entry hub, and renders three Asian warehouses (Chennai, Osaka, Shenzhen) economically unjustifiable. The Full Redesign closes these three facilities, reducing fixed warehouse costs from 22,046€ to 15,915€. Feasibility is maintained through four emergency airlift arcs towards Dubai (H2), carrying a combined 2,853 units at  $4.5\times$  the baseline corridor cost. These emergency flows represent the primary cost driver, pushing variable transport to 7,059,318€ — a +60% increase over baseline. Strategy A retains all 14 warehouses rather than pay closing penalties for Chennai, Osaka, and Shenzhen (whose combined opening cost is 6,131€), incurring 9,973 € more than F. Strategy R is further penalised by inflexible routing through the fixed baseline arc set.

**Table 13: Strategy comparison — S2 (Korea–Europe corridor blockade).**

| Strategy | Total (€) | $\Delta Z$ (€) | $\Delta Z$ (%) | Prem. vs F (€) |
|----------|-----------|----------------|----------------|----------------|
| F        | 4,454,801 | $\approx 0$    | 0.00%          | —              |
| A        | 4,454,801 | $\approx 0$    | 0.00%          | 0              |
| R        | 4,454,801 | $\approx 0$    | 0.00%          | 0              |

F=A=R=Z\*: blocked arcs are unused in baseline.

**S2 — Korea–Europe corridor blockade** ( $\Delta Z \approx 0$ ). The removal of eight Korea-to-Europe arcs (including A015 Seoul→Frankfurt) has no measurable impact on total cost. This result, though counter-intuitive, is a direct consequence of the baseline routing: Seoul routes its entire semiconductor output through Singapore, never through the direct Seoul to Frankfurt corridor. The blocked arcs are unused in the optimal baseline flow — and no scenario-induced cost pressure makes them worth activating. The network's apparent resilience to S2 is not inherent robustness, however, but rather a by-product of Singapore over-centralisation: the Korea–Europe direct corridor is dominated by the Singapore routing. Under Singapore hub closure (S1), these corridors cannot be reactivated since they are themselves removed by S2. The co-occurrence of S1 and S2 would therefore be more severe than either alone.

**Table 14: Strategy comparison — S3 (Combined shock: T2 + S1).**

| Strategy | Total (€) | $\Delta Z$ (€) | $\Delta Z$ (%) | Prem. vs F (€) |
|----------|-----------|----------------|----------------|----------------|
| F        | 9,746,049 | +5,291,249     | +118.78%       | —              |
| A        | 9,755,073 | +5,300,272     | +118.98%       | +9,024         |
| R        | 9,769,939 | +5,315,139     | +119.31%       | +23,890        |

**S3 — Combined shock** ( $\Delta Z = +118.78\%$ ). Scenario S3, the superposition of T2 and S1, is the worst-case event in the portfolio and produces the highest absolute disruption cost at 5,291,249€. The combination is super-additive in practice: the energy crisis alone contributes +474,045€ and the Singapore closure alone contributes +2,636,979€, yielding a naive additive benchmark of 3,111,025€. The actual joint penalty reaches +5,291,249€, an excess of +2,180,224€ (+70.1% above the naive sum). This amplification occurs because the



T2 energy surcharge is applied to a network that has already been forced into its most expensive routing configuration by S1: every arc in the restructured topology is simultaneously hit by higher energy costs, compounding non-linearly on top of the structural disruption. The penalty is consequently severe: semiconductors account for 7,282,133€ of the variable cost under S3-F, a +158.7% increase over the baseline, reflecting the full cost of rerouting all Asian semiconductor supply through Dubai under both emergency routing and elevated energy prices across all other corridors. The same four emergency arcs activated under S1 are required here, their per-unit cost is  $7.56\times$  the original baseline corridor rate, reflecting the  $\times 4.5$  structural air-freight premium compounded by the 5% Asia–Middle East interzonal tariff surcharge and the 60% tariff on air freight. Under S3, Full Redesign reduces the open warehouse set to 11 facilities (dropping Chennai, Osaka, and Shenzhen), while Adaptation retains all 14, incurring 9,024€ in excess fixed costs.

**Table 15: Strategy comparison — S4 (Persian Gulf / Strait of Hormuz crisis).**

| Strategy | Total (€) | $\Delta Z$ (€) | $\Delta Z$ (%) | Prem. vs F (€) |
|----------|-----------|----------------|----------------|----------------|
| F        | 4,701,942 | +247,141       | +5.55%         | —              |
| A        | 4,701,942 | +247,141       | +5.55%         | 0              |
| R        | 4,723,597 | +268,797       | +6.03%         | +21,655        |

F=A: Adaptation converges to the Full Redesign solution.

**S4 — Persian Gulf / Strait of Hormuz crisis** ( $\Delta Z = +5.55\%$ ). The 14-arc removal of Dubai’s maritime routes to the Americas, combined with a Suez surcharge (+25%) and a war-risk premium (+40%) on Dubai’s remaining active corridors, raises cost by 247,141€. The impact is moderate because Dubai’s primary role in the baseline network is to serve Europe and Asia — the Americas are predominantly supplied via Chicago (H4) and São Paulo (H5), whose routes are unaffected. F and A yield identical totals, indicating that starting from the baseline configuration and allowing full structural adjustments converges to the same optimal solution as a clean redesign. Rerouting (R) pays 21,655€ more, incurred through suboptimal routing constrained to the fixed baseline arc set.

## VI STRATEGIC CONCLUSIONS

**RQ3 — Most devastating scenario.** S3 (Combined shock) is unambiguously the most devastating scenario with  $\Delta Z = +5,291,249\text{€}$  (+118.78%). Its severity is not merely the sum of its components: it is a compound vulnerability amplifier. The closure of H3 Singapore — the sole entry hub for all semiconductor supply and a critical node for battery components — forces a complete rerouting of 2,853 units through emergency Dubai airlifts priced at  $4.5\times$  baseline. When those emergency arcs are themselves subject to an energy surcharge (T2), the marginal cost of each rerouted semiconductor unit escalates dramatically. Semiconductors’ variable cost reaches 7,282,133€ under S3-F, an increase of 4,467,392€ compared to the unstressed baseline: a +158.7%

increase on a product that already concentrates 75.0% of total variable cost. The structural cause is the absence of a cost-competitive alternative to Singapore for East Asian flows: Dubai is the only viable backup hub, but its routing to North American customers has been simultaneously degraded by the energy shock.

**RQ4 — Interdependent structural vulnerabilities.** Three interdependent vulnerabilities characterise the baseline configuration.

- The dominant one is H3 Singapore concentration. Scenarios S1 and S3 together represent 7,928,228€ in disruption cost (82.09% of the cumulative  $\sum \Delta Z$ ) and both stem from the removal of a single hub. Singapore serves as the sole entry point for all the semiconductors-related and battery-component supplier connections to the East Asian warehouse tier, and as the operational hub for six regional warehouses. This single-point-of-failure topology is the network’s primary systemic risk, perfectly illustrating how prioritizing pure cost-efficiency creates severe vulnerabilities that undermine supply chain resilience.
- The second vulnerability is arc A133 (Osaka W6 → Chicago H4), which carries  $\approx 536$  units of semiconductors in the baseline and represents the only viable bridge between the Asian warehouse tier and North American demand. Under S1/S3, Osaka closes and Osaka-Chicago arc disappears, its replacement (emergency airlifts at  $4.5\times$  cost through Dubai) is the largest single driver of the observed  $\Delta Z$ .
- The third vulnerability is structural over-opening: the baseline keeps 14 of 15 warehouses open (W9 Sydney alone is closed), with several facilities operating at marginal utility, such as Chennai (W5, 4.1% utilisation) and Monterrey (W13, 8.7%). Under Singapore hub disruption, the Asian facilities in this category (notably Chennai, along with Osaka and Shenzhen) lose their supply access and become pure fixed-cost liabilities that the Adaptation strategy cannot shed without penalty.

**RQ5 — Cost of resilience.** The formal computation of  $Z^*_{\text{robust}}$  operationalises the resilience investment identified in Section V-A. The robustified network is obtained by solving Phase 1 with four pre-contracted backup air arcs from Asian suppliers to Dubai (H2): 3 of them carry semiconductors from Taipei, Seoul, and Yokohama, while the fourth carries Battery Components from Chengdu. Each arc (4 in total) carries a fixed activation cost of 2,500€ at a standard pre-contracted rate, in contrast to the  $4.5\times$  emergency multiplier imposed when these routes are activated reactively. The resulting resilience premium is:

$$\begin{aligned}\Delta Z_{\text{resilience}} &= Z^*_{\text{robust}} - Z^* = 4,464,801 - 4,454,801 \\ &= 10,000 \quad (+0,224\%).\end{aligned}$$

The network topology is otherwise unchanged: the same 14 warehouses remain open and 11 baseline fixed arcs remain active. The four backup arcs are pre-contracted but carry zero

flow in normal operating conditions, the investment is purely a hedge against Singapore hub disruption.

The impact of this investment, evaluated across all seven disruption scenarios, is reported in Table 16.

**Table 16:** Disruption cost comparison: standard baseline vs. robustified network.  $\Delta Z$  values are relative to each network's own  $Z^*$ . All figures under the Full Redesign strategy.

| Sc.      | $\Delta Z$ (std, €) | $\Delta Z$ (rob, €) | Reduction (€)    | Red. (%) |
|----------|---------------------|---------------------|------------------|----------|
| T1       | +155,456            | -126,252            | 281,708          | 181.2    |
| T2       | +474,045            | +449,315            | 24,731           | 5.2      |
| T3       | +852,284            | +504,112            | 348,172          | 40.9     |
| S1       | +2,636,979          | +338,364            | 2,298,615        | 87.2     |
| S2       | $\approx 0$         | -239,591            | 239,591          | —        |
| S3       | +5,291,249          | +1,613,464          | 3,677,784        | 69.5     |
| S4       | +247,141            | +74,419             | 172,723          | 69.9     |
| $\Sigma$ | +9,657,154          | +2,613,830          | <b>7,043,324</b> | —        |

The primary effect is the near-elimination of the structural vulnerability to Singapore closure. Under S1,  $\Delta Z$  falls from 2,636,979€ to 338,364€ (−87.2%): the four pre-contracted arcs serve as the sole rerouting mechanism, at standard variable rates rather than the  $4.5\times$  emergency multiplier. The residual cost of 338,364€ reflects the genuine routing detour through Dubai and the loss of three Asian warehouses (Chennai, Osaka and Shenzhen) whose supply access is severed by Singapore closure, regardless of the backup capacity available. Under S3, the additional energy surcharge on the now-essential Dubai routes inflates the backup arc variable cost.

A structural side-effect of the pre-contracted Dubai arcs is visible across the tariff scenarios. Under T1, the robust network achieves  $\Delta Z = -126,252$ €: the +40% tariff on trans-oceanic corridors makes those routes sub-optimal, while the pre-contracted Asia–Dubai arcs (exempt from the tariff penalty) become cost-competitive enough that redirecting flow through them lowers total cost *below*  $Z^*_{\text{robust}}$ . The investment thus acts simultaneously as a disruption hedge and as a tariff arbitrage instrument. A similar logic applies in T2, T3, S2, and S4, where the backup arcs provide alternative routing capacity that buffers cost shocks extending well beyond the originally targeted Singapore failure mode.

The aggregate return on this investment is:

$$\begin{aligned} \text{ROI} &= \frac{\Sigma \Delta Z_{\text{std}} - \Sigma \Delta Z_{\text{rob}}}{\Delta Z_{\text{resilience}}} \\ &= \frac{9,657,154 - 2,613,830}{10,000} \approx 704.3\times \end{aligned}$$

Equivalently, the investment breaks even if the probability of at least one disruption scenario materialising exceeds 0.14%, a threshold that any realistic geopolitical risk assessment would exceed. This directly operationalises the theoretical trade-off between robustness and flexibility: a modest proactive investment in pre-contracted capacity (+0.224% of  $Z^*$ ) eliminates the need for reactive emergency routing at  $4.5\times$  cost, reducing the cumulative disruption burden by 7,043,324€ across the seven-scenario portfolio.

## VII CONCLUSION

This work has applied a Multi-Commodity Capacitated Fixed-Charge Network Design framework to the GlobalFlow logistics network, yielding four precise answers to the research questions asked.

**RQ1.** Among the four formulations, F0 produces the tightest LP relaxation (gap 0.5343%). The disaggregated per-arc warehouse linking constraint (Axis W) tightens the LP bound by individually bounding each incident arc to  $u_{ij} \cdot y_i$ , preventing the solver from saturating any arc while keeping the warehouse only partially open. Axis A disaggregation loosens the bound: the per-product constraint  $x_{ij}^p \leq u_{ij} \cdot z_{ij}$  allows a combined arc flow of  $|P| \cdot u_{ij} \cdot z_{ij}$ , three times more permissive than the aggregated form. F0 achieves the tightest relaxation by combining strong Axis W with tight Axis A (aggregated), confirming that both axes must be optimised jointly to maximise LP bound quality.

**RQ2.** The exact baseline optimum is  $Z^* = 4,454,801$ €, with 14 of 15 warehouses open and 11 of 47 fixed arcs activated. The cost structure is dominated at 99.1% by variable transport (fixed costs  $\leq 1\%$ ) with semiconductors concentrating 63.8% of variable cost despite accounting for only 44% of total flow.

**RQ3.** S3 (combined Singapore closure and energy crisis) is the most devastating scenario ( $\Delta Z = +5,291,249$ €, +118.78%). Its severity originates in the structural over-dependence on H3 Singapore as the sole entry hub for all semiconductor supply: scenarios S1 and S3 together account for 82.09% of the cumulative disruption cost  $\Sigma \Delta Z = 9,657,154$ €.

**RQ4.** The baseline network exhibits three interdependent structural vulnerabilities. First, H3 Singapore concentration: scenarios S1 and S3 together account for 7,928,228€ (82.09% of  $\Sigma \Delta Z$ ), both stemming from the removal of a single hub serving as the sole entry point for all semiconductor and battery-component flows into the East Asian warehouse tier. Second, arc A133 (Osaka W6  $\rightarrow$  Chicago H4) is the only viable bridge to North American demand, whose disappearance under S1/S3 forces emergency airlifts at a  $4.5\times$  cost multiplier. Third, structural over-opening leaves several facilities at marginal utilisation (Chennai W5 at 4.1%, Monterrey W13 at 8.7%), which become pure fixed-cost liabilities under Singapore disruption.

**RQ5.** Pre-contracting four supplier-to-Dubai backup arcs (3 for semiconductors; 1 for battery components) at 2,500€ each yields  $Z^*_{\text{robust}} = 4,464,801$ €, a resilience premium of exactly 10,000€ (+0.224%). This investment reduces S1 disruption cost by −87.2% (from 2,636,979€ to 338,364€) and S3 by −69.5% (from 5,291,249€ to 1,613,464€), by eliminating the  $4.5\times$  emergency multiplier in favour of pre-agreed standard rates. The cumulative disruption burden across all seven scenarios falls from 9,657,154€ to 2,613,830€ (−72.9%), yielding an aggregate ROI of 704.3× with a break-even disruption probability of 0.14%.